→ Height of a fuzzy set
$h(A)$

→ Normal fuzzy set
$A(x)$ as normal if $h(A)=1$

→ Sub normal fuzzy set
$A(x)$ is a subnormal if $h(A)<1$

Some standard operations in fuzzy sets.

→ Proper subset
$A(x)$ $B(x)$
$x \in X$
$A(x) \subset B(x)$, if $\mu_A(x) < \mu_B(x)$
$A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$
$B(x) = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0.8)\}$

→ Equal fuzzy set
$A(x) = B(x)$
$A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$
$B(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$

→ Compliment of a fuzzy set

$A(x)$ $\quad \bar{A}(x)$

$\bar{A}(x) = 1 - A(x)$

If $A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$

$\bar{A}(x) = \{(x_1, 0.9), (x_2, 0.8), (x_3, 0.7)\}$

→ Intersection of fuzzy set:

$A(x)$ $\quad B(x)$ $\quad x \in X$

$(A \cap B)(x)$

$\mu_{(A \cap B)}(x) = \min\{\mu_A(x), \mu_B(x)\}$

$A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$

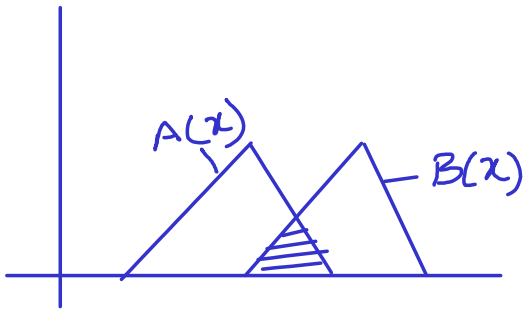
$B(x) = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0.8)\}$

$\mu_{(A \cap B)} x_1 = \min\{\mu_A(x_1), \mu_B(x_1)\} = 0.1$

$\mu_{(A \cap B)} x_2 = 0.2$

$\mu_{(A \cap B)} x_3 = 0.3$

$$(A \cap B)(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$$



→ Union of fuzzy sets

$$\mu_{(A \cup B)}(x) = \max\{\mu_A(x), \mu_B(x)\}$$

→ Algebraic product of fuzzy sets.

Algebraic product of two fuzzy sets $A(x)$ and $B(x)$ is denoted by $A(x) \cdot B(x)$

$$A(x) \cdot B(x) = \{x, \mu_A(x) \cdot \mu_B(x)), x \in X\}$$

Ex: $A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$

$B(x) = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0.8)\}$

$$A(x) \cdot B(x) = \{(x_1, 0.05), (x_2, 0.14), (x_3, 0.24)\}$$

→ Multiplication of fuzzy set with a crisp value:

$$d \cdot A(x) = \{ (x, d \times \mu_A(x)), x \in A \}$$

Ex: $A(x) = \{ (x_1, 0.1), (x_2, 0.2), (x_3, 0.3) \}$

If $d = 2$

$$d \cdot A(x) = \{ (x_1, 0.2), (x_2, 0.4), (x_3, 0.6) \}$$

→ Algebraic sum of two fuzzy sets:

$A(x)$ $B(x)$ $x \in X$

$$A(x) + B(x) = \{ (x, \mu_{A+B}(x)), x \in X \}$$

$$\mu_{A+B}(x) = \mu_A(x) + \mu_B(x) - \mu_A(x) \cdot \mu_B(x)$$

$$A(x) = \{ (x_1, 0.1), (x_2, 0.2), (x_3, 0.3) \}$$

$$B(x) = \{ (x_1, 0.5), (x_2, 0.7), (x_3, 0.8) \}$$

$$A(x) + B(x) = \{ (x_1, 0.55), (x_2, 0.76), (x_3, 0.86) \}$$

→ Bounded sum of two fuzzy set $A(x)$ and $B(x)$

Bounded sum is denoted as follows:

$$A(x) \oplus B(x) = \{(x, \mu_{A \oplus B}(x)), x \in X\}$$

Where $\mu_{A \oplus B}(x) = \min\{1, \mu_A(x) + \mu_B(x)\}$

Example:

$$A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$$

$$B(x) = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0.8)\}$$

$$A(x) \oplus B(x) = \{(x_1, 0.6), (x_2, 0.9), (x_3, 1)\}$$

→ Algebraic difference of two fuzzy sets $A(x)$ and $B(x)$ is represented as follows:

$$A(x) - B(x) = \{(x, \mu_{A-B}(x)), x \in X\}$$

$$\mu_{A-B}(x) = \mu_{A \cap \bar{B}}(x)$$

$$A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$$

$$B(x) = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0.8)\}$$

$$\bar{B}(x) = \{(x_1, 0.5), (x_2, 0.3), (x_3, 0.2)\}$$

$$A(x) - B(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.2)\}$$

→ Bounded difference of two fuzzy sets $A(x)$ and $B(x)$ is represented as follows:

$$A(x) \ominus B(x) = \{(x, \mu_{A \ominus B}(x), x \in X\}$$

$$\mu_{A \ominus B} = \max\{0, \mu_A(x) + \mu_B(x) - 1\}$$

Example:

$$A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$$

$$B(x) = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0.8)\}$$

$$A(x) \ominus B(x) = \{(x_1, 0), (x_2, 0), (x_3, 0.1)\}$$

→ Cartesian product of two fuzzy sets $A(x)$ and $B(y)$ is represented as $A(x) \times B(y)$

$A(x)$ $\quad x \in X$

$B(y)$ $\quad y \in X$

$$\mu_{A \times B}(x, y) = \min(\mu_A(x), \mu_B(y))$$

$$A(x) = \{(x_1, 0.2), (x_2, 0.3), (x_3, 0.5), (x_4, 0.6)\}$$

$$B(y) = \{(y_1, 0.8), (y_2, 0.6), (y_3, 0.3)\}$$

$$\min(x_1, y_1) = \min(0.2, 0.8) = 0.2$$

$$A \times B = \begin{bmatrix} 0.2 & 0.2 & 0.2 \\ 0.3 & 0.3 & 0.3 \\ 0.5 & 0.5 & 0.3 \\ 0.6 & 0.6 & 0.3 \end{bmatrix}$$